Inspection record: Polarization (Griffiths §4.1)

Summary of the study notes for book pp. 166–172 · Electrodynamics II, Module I

SEM5-EM2-M1-4.1-SIR-01
Rev A · 2026-10-14

Subject	Electric polarization of dielectrics: induced dipoles, permanent dipoles, polarization P
Source reviewed	Griffiths, *Introduction to Electrodynamics* (5th ed.), §4.1, pp. 166–172; study notes “polarization-study-notes”
Scope	Concepts, derivations and units of §4.1 as written in the study notes. Not §4.2 onward.
Method	Each derivation re-worked by hand from the base result. Units checked dimensionally. Book equation numbers cross-checked against the excerpt.
Prepared by	Student (self-review)

ITEMS INSPECTED	ACCEPTED	NOTES	DISPOSITION
13	**7**	**3**	**ACCEPTED WITH OPEN ITEMS (3)**

A. INSPECTION ITEMS

Item	Inspected concept	Acceptance criterion	Finding	Status
I-01	Conductor vs dielectric	Conduction electrons move through the body; dielectric charges are bound to atoms or molecules.	Consistent with book §4.1.1.	ACCEPT
I-02	Induced dipole model (Ex. 4.1)	Equilibrium at $E = E_e$ gives $p = qd$ and $p \propto E$ (Eq. 4.1).	Re-derived: $p = (4\pi\epsilon_0 a^3)E$.	ACCEPT
I-03	Field of uniform charged ball	For $s < a$, enclosed charge is Qs^3/a^3; Gauss law gives $E = Qs/(4\pi\epsilon_0 a^3)$.	Re-derived. Matches the step in Ex. 4.1.	ACCEPT
I-04	Polarizability in volume form	$\alpha = 4\pi\epsilon_0 a^3 = 3\epsilon_0 v$ (Eq. 4.2), using $v = \frac{4}{3}\pi a^3$.	Algebra checked.	ACCEPT
I-05	Units of α; Table 4.1	α in C²·m/N; $\alpha/4\pi\epsilon_0$ in $10^{-30}\ \mathrm{m}^3$.	Dimensional check agrees for both.	ACCEPT
I-06	Molecular polarizability	Direction-dependent α; 3×3 tensor (Eq. 4.3).	Summarised only. Eq. 4.3 not derived in full.	NOTE
I-07	Torque on permanent dipole	Net force zero; torque $\mathbf{p} \times \mathbf{E}$ (Eq. 4.4).	Re-derived from the two end forces about the centre.	ACCEPT
I-08	Potential energy $U = -\mathbf{p} \cdot \mathbf{E}$ (Eq. 4.6)	Work integral from $90°$ gives $-pE\cos\theta$.	Re-derived. Valid for permanent dipoles only (book fn. 3).	NOTE
I-09	Force in non-uniform field (Eq. 4.5)	$\mathbf{F} = (\mathbf{p} \cdot \nabla)\mathbf{E}$ from first-order difference of $\mathbf{E}$ across $\mathbf{d}$.	Outline only. Partial derivatives not yet confirmed as learned.	OPEN
I-10	Polarization $\mathbf{P} = N\mathbf{p}$	Dipole moment per unit volume; unit C/m².	Valid for aligned identical dipoles.	ACCEPT
I-11	Thermal alignment	Compare pE with $k_B T$; alignment weak when $pE \ll k_B T$.	Qualitative only. No Boltzmann estimate done.	NOTE

I-12	“Factor of about 4” accuracy of Eq. 4.2	Claim checked against the book or a reference value.	Not verified. Source of the claim not checked.	OPEN
I-13	Problem 4.2 (hydrogen cloud)	Expansion $e^{-2r/a} \approx 1 - 2r/a$ gives α for $\rho \propto e^{-2r/a}$.	Idea stated; calculation not carried out.	OPEN

B. OPEN ITEMS AND REQUIRED ACTIONS

Item	Action required before the summary is final	Reason
I-09	Revise partial derivatives and the gradient ∇; then recheck Eq. 4.5 step by step.	Eq. 4.5 depends on ∇, which is not in Plus Two.
I-12	Check the factor-of-4 claim against the book text or a source. Do not quote it until confirmed.	Stated from memory in the notes; not derived.
I-13	Carry out the Problem 4.2 integral: enclosed charge for $\rho \propto e^{-2r/a}$, then the field, then α.	The notes state the idea but do not finish the calculation.

C. LIMITATIONS OF THIS RECORD

This is a self-review study record. It is not an instructor-approved or certified document, and it is not a maintenance or airworthiness record. The figures in the study notes are my own sketches, not the book’s figures, and they have not been checked against the book’s page images. Items I-06, I-08, I-11 are summarised, not fully derived.

D. SIGN-OFF

Inspected by (name)		Date	
Reviewed by (instructor)		Date	

Study inspection record. Equation numbers refer to Griffiths 5th ed. §4.1. Status key: ACCEPT = criterion met; NOTE = met with a stated limit; OPEN = not yet closed.